

Section 4.3

I. Definition: A function f is said to be *strictly increasing* on an interval I if

$$x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$$

Definition: A function f is said to be *strictly decreasing* on an interval I if

$$x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$$

Definition: A function f is said to be *strictly monotone* on an interval I if f is either strictly increasing or strictly decreasing on I

II. **Montone Function Theorem** Let f be differentiable on the open interval (a, b) .

a. If $f'(x) > 0$ on (a, b) , then f is strictly increasing on (a, b) .

b. If $f'(x) < 0$ on (a, b) , then f is strictly decreasing on (a, b) .

Examples

III. First Derivative Test for Relative Extrema

Step 1. Find all critical number of f

Step 2. Classify each critical point $(c, f(c))$

a. Relative maximum

b. Relative minimum

c. Neither

Examples

IV. Definition: A function f is said to be *concave up* on an interval I if

the graph of f lies above all of its tangents on the interval I

Definition: A function f is said to be *concave down* on an interval I if

the graph of f lies below all of its tangents on the interval I

Definition: A point $P = (c, f(c))$ on the graph of f is said to be an *inflection point* of f if

the graph of f is concave up on one side of P and concave down on the other side of P

V. **Concavity Theorem** Let f be differentiable on the open interval (a, b) .

a. If $f''(x) > 0$ on (a, b) , then f is concave up on (a, b) .

b. If $f''(x) < 0$ on (a, b) , then f is concave down on (a, b) .

Examples

VI. Second-Derivative Test for Relative Extrema

Step 1. Let c be a critical number of f such that $f'(c) = 0$

Step 2. Classify the critical point $(c, f(c))$

a. If $f''(c) > 0$, then f has a relative minimum at $x = c$

b. If $f''(c) < 0$, then f has a relative maximum at $x = c$

c. If $f''(c)$ vanishes, then the test **fails**

Examples